

ORIGINAL ARTICLE

Consensus Guideline on the Management of Epilepsy in Egypt: A National Delphi Consensus Study

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Abstract

Objective: In epilepsy, early diagnosis, accurate determination of epilepsy type, proper selection of antiseizure medication, and monitoring are all essential. However, despite recent therapeutic advances and conceptual reconsiderations in the classification and management of epilepsy, serious gaps are still encountered in day-to-day practice in Egypt as well as several other resource-limited countries. Premature mortality, poor quality of life, socio-economic burden, cognitive problems, poor treatment outcomes, and comorbidities are major challenges that require urgent actions to be implemented at all levels. In recognition of this, a group of Egyptian epilepsy experts met through a series of consecutive meetings to specify the main concepts concerning the diagnosis and management of epilepsy, with the ultimate goal of establishing a nationwide Egyptian consensus.

Methods: The consensus was developed through a modified Delphi methodology. A thorough review of the most recent relevant literature and international guidelines was performed to evaluate their applicability to the Egyptian situation. Afterward, several remote and live rounds were scheduled to reach a final agreement for all listed statements.

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Results: Of 278 statements reviewed in the first round, 256 achieved $\geq 80\%$ agreement. Live discussion and refinement of the 22 statements that did not reach consensus during the first round took place, followed by final live voting then consensus was achieved for all remaining statements.

Significance: With the implementation of these unified recommendations, we believe this will bring about substantial improvements in both the quality of care and treatment outcomes for persons with epilepsy in Egypt.

Plain Language Summary: This work represents the efforts of a group of medical experts to reach an agreement on the best medical practice related to people with epilepsy based on previously published recommendations while taking into consideration applicable options in resource-limited countries. The publication of this document is expected to minimize many malpractice issues and pave the way for better healthcare services on both individual and governmental levels.

KEYWORDS

antiseizure medication, consensus, Delphi method, drug resistant epilepsy, Egypt, epilepsy, epilepsy diagnosis, epilepsy surgery, status epilepticus, Steering Committee

1 | INTRODUCTION

Epilepsy is a chronic disorder of the brain that is marked by episodes of seizures. It arises from various pathological insults and is considered heterogeneous rather than a specific disease.¹ The prevalence of epilepsy is around 50 million people worldwide, per the World Health Organization (WHO) figures, and most (almost 80%) reside in low- and middle-income countries.² Moreover, the premature mortality risk is almost threefold in persons with epilepsy (PWE) compared to the public. The prevalence of epilepsy in Egypt was estimated to be 6.98/1000.³ The lifetime prevalence of epilepsy in children and adolescents in upper Egypt was estimated to be 9.7/1000, with a lower prevalence among adolescents than children below 12 years (7.2/1000 and 10.8/1000), respectively.¹

Despite epilepsy being one of the oldest illnesses in history,⁴ serious difficulties and challenges are still encountered across the entire patient journey. Establishing the definitive epilepsy diagnosis and detecting the root cause are challenging tasks, particularly where cultural constraints hinder disease acceptance.⁵ Unlike other neurological disorders, epilepsy can develop due to different neuropathological alterations rather than a standalone disease.⁶ One of five patients treated for epilepsy does not really have the disease. Furthermore, the number of available antiseizure medications (ASMs) has been notable in recent years, making the proper treatment choice more

Key points

- Many published guidelines, although scientifically efficient, their application in developing countries may be limited, thus establishing a revised versions for regions with limited-resource countries will be more appropriate.
- Adapting consensus to uniform the diagnosis and management of epilepsy in developing countries will inevitably improve the care for people with epilepsy in these regions.
- The established consensus is expected also to decrease the burden of healthcare services on both individual and governmental levels.

complex.⁷ According to the WHO data, proper diagnosis and treatment can provide seizure freedom for 70% of PWE.² The remaining 30% present a major challenge due to resistance to drugs.⁶

As in other parts of the world, inadequacies in the care of PWE, such as misdiagnosis, inappropriate medication, and sudden unexpected death, are all well recognized. We sincerely hope this can be changed with the emergence of this national consensus. With all the efforts that this initiative bears, it represents a series of efforts devoted to creating a significant impact on the health of Egyptians

in recent years. We believe implementing these structured recommendations will substantially improve the quality of care and treatment outcomes for PWE across all age groups (children and adults) in Egypt. This gathering aims to prepare a consensus/recommendation from Egyptian epilepsy experts about the definition, diagnosis, and management of epilepsy within different age groups in Egypt.

2 | METHODS

Using a modified Delphi rotational technique, we used a competency-based approach to epilepsy curriculum development to *guide, uniform, and optimize diagnosis and management* by supporting all healthcare providers with essential information.

We designed a curriculum competency study to examine a large competency dataset and its applicability to develop an epilepsy curriculum that suits a limited-resource country such as Egypt. The board was composed of recognized, highly specialized experts in epilepsy; 34 Egyptian experts, including 30 neurologists and 4 neurosurgeons representing 14 Egyptian university hospitals and healthcare institutions.

Through a preliminary meeting, *four main sections* were identified to constitute the main core for this consensus' recommendation, namely: (1) definition, classification, and diagnosis of epilepsy, (2) choice of ASMs, (3) management of status epilepticus, and (4) management of drug-resistant epilepsy (DRE). The four sections were designed to cover all aspects of epilepsy's main practice.

According to these four sections, *preselected items* were drawn from various *sources*, including related competency profiles, interviews with selected content experts, and synthesized literature reviews conducted across different databases, including Web of Science, Medline, PubMed, and the Cochrane Library. Keywords during the search were "guidelines," "epilepsy management," "epilepsy diagnosis," "DRE," and "Egypt," and we selected guidelines papers and excluded others.

These preselected items were extensively discussed and rephrased to address the preliminary statements under each section. Finally, constructed and carefully selected *items'* statements were prepared as intensive closed-ended questionnaires interspersed with controlled-opinion feedback. Each item was to be rotated to the panelists for at least one round. The two main components of the study were the procedure itself (i.e., a series of rounds with selected experts) and intent (i.e., to predict future events and to arrive at an accepted consensus).

We continuously worked to reduce the *effects of bias* due to group interaction, assuring anonymity and providing controlled feedback. We used *subpanels* to reduce the level of fatigue on panelists and increase the volume of competencies that can be effectively and efficiently analyzed.

During the first round in July 2020, each panelist anonymously voted on each item's statement using a 3-point scale (1 = agree, 2 = neutral, and 3 = disagree). For each item's statement, a consensus was considered achieved if 80% or more of the members voted with an agreement. Of 278, only 22 items' statements did not achieve the consensus agreement and required a second round.

During the second round in April 2021, live discussion and refinement of the 22 statements that did not reach consensus during the first round took place, followed by final live voting then consensus was achieved for all remaining statements (Figure 1): Consensus development by the modified Delphi methodology.

In our manuscript, we defined conflicts of interest as any financial or personal interest that could influence the interpretation or presentation of our research findings. We disclosed all potential conflicts of interest in the manuscript submission process. We also followed the guidelines of the journal for disclosing conflicts of interest,

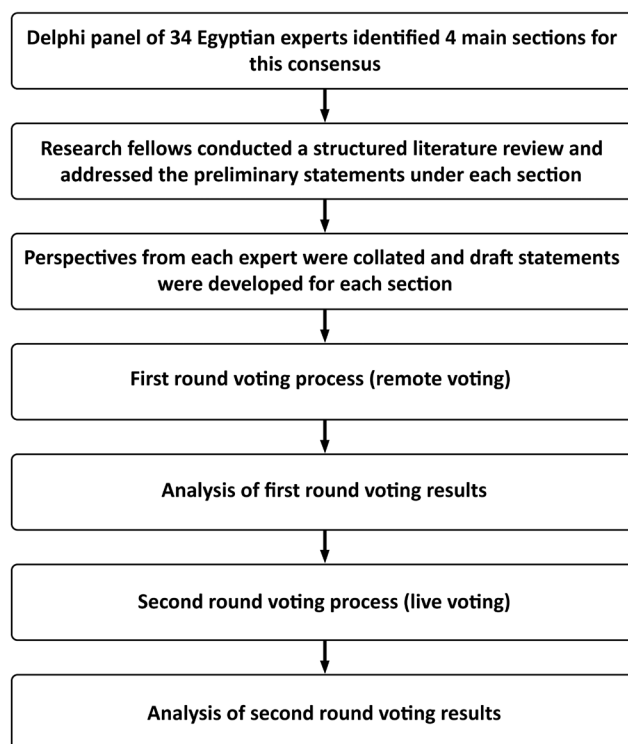


FIGURE 1 Consensus development by the modified Delphi methodology.

which includes reporting any financial relationships or potential conflicts of interest related to the research, as well as any other relevant financial or nonfinancial relationships that could be perceived as a conflict of interest. Furthermore, all authors declare that they have no financial relationships or interests that could be perceived as a conflict of interest with any commercial entity mentioned in the manuscript.

3 | RESULTS

3.1 | Section I: Definition, classification, and diagnosis of epilepsy

3.1.1 | Definition and classification of epilepsy

The Steering Committee (SC) agreed to adopt the International League Against Epilepsy (ILAE) 2014 definition of epilepsy and the 2017 classification of epilepsies and seizures.^{8–10} The full voting results are presented in the [Tables S1–S23](#).

3.1.2 | General outlines for epilepsy diagnosis

In acknowledgement of certain challenges that are associated with the proper diagnosis of people with epilepsy, the collaboration with other specialties such as pediatricians in cases of neonatal seizures and neuro-metabolic disorders and psychiatrists in patients with psychiatric morbidities is highly endorsed. In view of these considerations, the SC agreed to adopt the following recommendations:

1. To determine whether the patient might have had an epileptic seizure, the patient's and eyewitness's (wherever possible) detailed history should be taken.
2. A clinical and neurologic examination is crucial since recurrence can be predicted through an abnormal examination after a first seizure.
3. In children and young people, a specialist neurologist with training and expertise in epilepsy should establish the diagnosis of epilepsy.
4. Healthcare professionals should enable PWE and their caregivers to be involved in all decisions as partners. Furthermore, their culture, race, gender, and specific needs should be considered.
5. If the epilepsy diagnosis cannot be definitely confirmed, further investigations are recommended, and/or referral to a more specialized epilepsy center should be done.

6. Where psychogenic nonepileptic seizures (PNES) are confirmed, suitable referrals to psychiatric services should be made for further investigation and treatment.

3.1.3 | Electroencephalography for epilepsy diagnosis

The SC agreed that electroencephalography (EEG) is a valuable tool for confirming and classifying seizure disorders. The SC highlighted the below recommendations:

1. A normal EEG^a should not exclude the diagnosis of epilepsy.
2. Epilepsy is a clinical diagnosis; however, EEG can be indicated in certain^b conditions.
3. When there is clinical doubt, EEG and/or ictal EEG should be used to classify epileptic seizures and epilepsy syndromes.
4. In case of equivocal epilepsy diagnosis or suspected nonepileptic events, short-term video-EEG should be available (with attempts to provoke such events as by suggestion) to support the diagnosis.
5. In patients with diagnostic difficulties, inpatient video-EEG monitoring and other special investigations (including polysomnography with full EEG montages) should be available to support the diagnosis.
6. When findings of a standard EEG have not supported epilepsy diagnosis or classification, a sleep EEG should be performed.
7. The test should not be performed in patients with a typical history of other disorders (such as migraine and syncope) since the probability of epilepsy will be minor.
8. Increasing recording time to 60 min on conventional EEG is a more convenient and cost-effective method of enhancing the EEG diagnostic yield compared to multiple conventional EEGs.
9. In the case of patients with a likelihood of a non-convulsive epileptic seizure, as suggested by the clinical history, the finding of epileptiform abnormalities is specific to assess the risk of seizure recurrence.
10. Video-EEG is the gold standard for the diagnosis of epilepsy and its mimics.

^bElectrical status epilepticus during sleep (ESES) and Non-convulsive status epilepticus (NCSE).

^aThroughout the consensus report, when electroencephalography "EEG" is mentioned, it indicates interictal EEG while if ictal EEG is intended the word "ictal" will always precede the acronym EEG

11. Recording attacks in the patient's usual setting is an advantage of ambulatory EEG recording, but it is still inconvenient and costly.
12. All 21 electrodes and placements should be used (10–20 systems) as the International Federation of Clinical Neurophysiology recommended.
13. Periods when the eyes are closed and open should be included in the recordings.
14. Photoc stimulation, sleep deprivation, and hyperventilation can be used routinely (unless for medical or other justifiable reasons).

3.1.4 | Electrocardiography for epilepsy diagnosis

The SC agreed to adopt the following recommendations:

1. In all patients with blackouts, the 12-lead electrocardiography (ECG) has great importance in the assessment of those patients, especially in older age groups (since epilepsy can be stimulated by abnormal heart rhythm).
2. ECG must be a routine channel during EEG recording and should be performed as a baseline before starting ASM.

3.1.5 | Magnetic resonance imaging (MRI)

The SC recognizes the important role of neuroimaging in epilepsy. Imaging modalities include but are not limited to brain CT, MRI, PET scan, SPECT, fMRI, and Wada test. In this section, consensus regarding standard imaging as CT and MRI is discussed while more advanced imaging modalities will be mentioned in upcoming sections related to their specific indications.

Consensus to adopt the following recommendations are as follows:

1. MRI epilepsy protocol should follow the ILAE consensus on the “recommendations for the use of structural MRI in the care of PWE”.¹¹
2. MRI is mandatory for brain imaging in PWE (before 2 years or in adults).
3. Any patient with focal onset on history, EEG, or examination (unless clear evidence of benign focal epilepsy) is indicated for MRI.
4. Any patient with seizures that continue despite first-line ASM is indicated for MRI.
5. Loss of seizure control with ASMs or variation in the pattern of seizure that may suggest a progressive underlying lesion is an indication for MRI.
6. When the diagnosis of genetically generalized epilepsy (GGE) is confirmed, there is no necessity for routine

brain imaging. However, in case of suspicion of another diagnosis, an MRI epilepsy protocol is advised.

7. Computed tomography (CT) has a role when there is an urgent need for assessment of seizures or when MRI is contraindicated.

3.1.6 | Genetic testing

New technologies for detecting different types of mutations have substantially improved understanding of epilepsy genetics, thus, allowing some patients for more efficient genetic diagnosis. This new approach can improve patient care, enhance prognosis, and support treatment decisions.¹² The SC agreed upon the following recommendations:

1. Expert advice on the genetics of epilepsy is still not available in Egypt.
2. For an accurate epilepsy syndromic diagnosis, a family history should be taken. Family history is also important to determine the risk of epilepsy development in children of parents with epilepsy.

3.1.7 | Neuropsychological assessment

The SC consensus upon adopting the following recommendations:

1. Access to neuropsychological specialist advice should be available as deemed appropriate by the multidisciplinary team.
2. Referral for a neuropsychological assessment is indicated for:
 - (i) Any PWE who is suffering from learning or occupational challenges. Memory problems should be identified by a cognitive screen in the first instance since they can be distressing and/or disabling.
 - (ii) It is indicated as a part of the presurgical evaluation.

3.1.8 | Laboratory studies

The SC agreed to adopt the following recommendations:

1. In adults, appropriate blood tests should be considered (complete blood count (CBC), glucose, calcium, thyroid-stimulating hormone (TSH), and plasma electrolytes) to specify possible causes and/or any significant comorbidity.
2. In adolescents and adults, urine toxicology screening should be done according to individual considerations and circumstances.

3. Consider lumbar puncture (LP) for a child <6 months, a patient who fails to return to baseline/meningeal signs or suspects high intracranial pressure (imaging before LP is mandatory).
4. In children and young people, other investigations should be considered in special situations (including urine and blood biochemistry) to identify an underlying cause of epilepsy and exclude other diagnoses.

3.2 | Section II: Choice of antiseizure medication

During our synthesized literature review, we identified seven guideline reports addressing ASM in epilepsy,^{7,13–18} two clinical trials,^{19,20} six review articles,^{21–26} and one survey.²⁷ We also recommend the referral to randomized trials^{28,29} and meta-analysis^{30,31} comparing different ASMs as regards type of seizures, clinical effectiveness, and cost-effectiveness to aid in the choice of ASM on individual basis on patient characteristics and drug availability. A total of 6 general considerations and 21 final statement recommendations regarding choice of ASM were submitted for assessment (Tables 1–3). We aimed to develop evidence-based treatments similar to approaches commonly used, with the intention of minimizing variations in treatment plans.

3.2.1 | General considerations

The SC agreed to adopt the following recommendations:

1. In case of failure of the initial ASM (due to continued seizures or adverse effects), initiation of an additional drug should take place and then escalate up to an adequate or maximum tolerated dose (this drug may be an alternative first-line or second-line drug). Afterward, slow tapering of the first drug should be done. Caution is needed during the changeover period.
2. If the second drug is worthless, tapering it or the first drug should be considered (this should be implemented considering efficacy, side effects, and tolerability of the drugs before starting another drug).
3. No available evidence shows whether alternative substitution or add-on therapy is more effective as a treatment strategy in the setting where the frequency of seizures after first medication is reduced by less than 50% or occurs at intervals less than threefold the frequency of the original seizure.
4. A considerable proportion of patients who do not achieve seizure freedom on monotherapy can benefit from combination therapy with the newer antiseizure medications.

5. The decision to discontinue antiseizure medication requires an individualized risk–benefit assessment by an epileptologist. In addition, the decision must be discussed with PWE and the patient's family/caregiver with an explanation of all the benefits and risks of medication discontinuation.
6. Drug withdrawal can be considered after a minimum of 2–5 years without a seizure based on the risk of seizure recurrence. Gradual withdrawal over 2–6 months is recommended. If seizures recur, the effective, well-tolerated drug previously used should be restarted using the last effective dose.

3.2.2 | Choice of ASM according to different epileptic syndromes in different age groups (after excluding treatable metabolic causes)

The SC agreed to adopt the recommendations depicted in Table 1.

3.2.3 | Choice of ASM according to different epileptic syndromes in adolescents

The SC agreed to adopt the recommendations depicted in Table 2.

3.2.4 | Choice of ASM according to seizure type

The SC agreed to adopt the recommendations depicted in Table 3.

3.2.5 | The use of brand versus generic drugs

The SC agreed to adopt the following recommendation:

1. A “brand” version of an ASM is not superior to a generic one. What is important is finding a version of a drug that suits the patient and is taken consistently. “Consistency of supply” means getting the same drug version with every prescription.

3.3 | Section III: Status epilepticus

Status epilepticus (SE) is “a neurologic emergency characterized by prolonged seizure activity or multiple seizures without return to baseline, with substantial morbidity and

TABLE 1 Recommendations for choice of antiseizure medication (ASM) according to different epileptic syndromes in different age groups (after excluding treatable metabolic causes).

No.	Recommendations	Contraindicated medications
<i>Self-limited epilepsy with centrotemporal spikes</i> ¹³		
1.	First line: carbamazepine, lamotrigine, levetiracetam, oxcarbazepine, and valproate. Adjuvant therapy: eslicarbazepine, lacosamide, lamotrigine, levetiracetam, and zonisamide. Other options: carbamazepine, clobazam, gabapentin, lamotrigine, levetiracetam, oxcarbazepine, pregabalin, tiagabine, topiramate, and vigabatrin.	-
<i>Self-limited epilepsy with autonomic seizures</i> ¹³		
2.	First line: carbamazepine and sodium valproate. Adjuvant therapy: eslicarbazepine, lacosamide, lamotrigine, levetiracetam, and zonisamide. Other options: carbamazepine, clobazam, gabapentin, lamotrigine, levetiracetam, oxcarbazepine, pregabalin, tiagabine, topiramate, and vigabatrin.	-
<i>Childhood occipital visual epilepsy</i> ¹³		
3.	First line: carbamazepine, lamotrigine, levetiracetam, oxcarbazepine, and valproate. Adjuvant therapy: eslicarbazepine, lacosamide, lamotrigine, levetiracetam, and zonisamide. Other options: carbamazepine, clobazam, gabapentin, lamotrigine, levetiracetam, oxcarbazepine, pregabalin, tiagabine, topiramate, and vigabatrin.	-
<i>Photosensitive occipital lobe epilepsy</i> ¹⁹		
4.	First line: avoid factors that provoke seizures and valproate. Adjuvant therapy: benzodiazepines and levetiracetam. Other options: carbamazepine, ethosuximide, lamotrigine, and vigabatrin.	-
<i>Childhood absence epilepsy</i> ¹³		
5.	First line: ethosuximide and valproic acid. Adjuvant therapy: a combination of ethosuximide and valproic acid. Other options: acetazolamide, clobazam, clonazepam, lamotrigine, levetiracetam, topiramate, and zonisamide.	Carbamazepine, oxcarbazepine, phenobarbital, phenytoin, tiagabine, and vigabatrin.
<i>Epilepsy with myoclonic absence</i> ²¹		
6.	First line: ethosuximide and valproic acid. Adjuvant therapy: a combination of ethosuximide and valproic acid. Other options: benzodiazepines, lamotrigine, phenobarbitone, and rufinamide.	Carbamazepine, phenytoin, and vigabatrin.
<i>Epilepsy with eyelid myoclonia</i> ²⁰		
7.	First line: ethosuximide and valproic acid. Adjuvant therapy: clonazepam, ethosuximide, and valproic acid. Other options: levetiracetam.	Carbamazepine, lamotrigine, oxcarbazepine, and phenytoin.
<i>Myoclonic atonic epilepsy</i> ²⁷		
8.	First line: valproate in generalized tonic-clonic seizure, ethosuximide in absence seizures, steroids in the periods with multiple atonic seizures, and frequent prolonged episodes of nonconvulsive status. Adjuvant therapy: lamotrigine and ketogenic diet. Other options: clobazam, felbamate, lamotrigine, levetiracetam, rufinamide, topiramate, zonisamide, and ketogenic diet.	Carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, and vigabatrin.
<i>Lennox–Gastaut syndrome</i> ¹³		
9.	First line: lamotrigine, topiramate, and valproate. Adjuvant therapy: lamotrigine, topiramate, and valproate. Other options: felbamate, rufinamide, and topiramate.	Carbamazepine, gabapentin, oxcarbazepine, pregabalin, tiagabine, and vigabatrin.

(Continues)

TABLE 1 (Continued)

No.	Recommendations	Contraindicated medications
<i>Developmental and/or epileptic encephalopathy with spike-wave activation during sleep</i> ²²		
10.	First line: standard ASMs, valproate, benzodiazepines, ethosuximide, adrenocorticotrophic hormone (ACTH) or prednisone, high-dose benzodiazepines, intravenous immunoglobulins, and epilepsy surgery. Adjuvant therapy: ----- Other options: acetazolamide, clobazam, lacosamide, lamotrigine, and levetiracetam.	-
<i>Febrile infection-related epilepsy syndrome</i> ²³		
11.	First line: benzodiazepines and barbiturates for treatment of the acute event. Adjuvant therapy: ketogenic diet. Other options: IVIG, cannabidiol, anakinra, and immunomodulation, such as tocilizumab or canakinumab, epilepsy surgery.	-
<i>Hemiconvulsion, hemiplegia epilepsy syndrome</i> ^{24,25}		
12.	First line: steroids, and N-methyl-D-aspartate (NMDA) receptor blockers such as memantine and amantadine. Adjuvant therapy; carbamazepine, and phenytoin in cases of persistent seizures. Other options; lamotrigine, perampanel, rufinamide, topiramate, valproate, and epilepsy surgery.	-

mortality rates”.³² The SC consensus is to adopt the following recommendations for the definition, diagnosis, evaluation, and management of SE (Figure 2): Status epilepticus management algorithm.

3.3.1 | Definition

The SC agreed to adopt the following definition:

1. SE is defined as “a seizure with 5 min or more of continuous clinical and/or convulsive seizure activity or recurrent seizure activity without recovery of consciousness between seizures.”³³

3.3.2 | Diagnosis and evaluation of SE

The SC agreed to adopt the following recommendations:

1. The diagnostic evaluation begins in parallel with emergent initial therapy.
2. For diagnosis, finger stick glucose should be checked, and pulse oximeter and cardiac monitoring should be started as soon as possible.
3. IV diazepam should be preferentially used if IV access is already established.
4. In case of the absence of IV access, intramuscular midazolam should be given as soon as possible.
5. Blood and serum laboratory evaluation typically includes CBC, basic metabolic panel, calcium, and magnesium determinations.

6. Some laboratory studies may be helpful in several patients, such as liver enzymes, troponin, blood gas determinations, toxicology screens, pregnancy tests, and ASM drug levels.
7. If a myocardial injury or cardiac arrhythmia is doubtful, ECG should be performed as far as possible.
8. Noncontrast CT of the brain is the first imaging study to be considered in the Emergency Department (ED) if MRI is not feasible.
9. In the case of febrile patients or if there is a suspected subarachnoid hemorrhage or central nervous system infection, LP should be performed, preferably after obtaining the CT scan.
10. In case the patient does not return within 30 min to the normal consciousness level, EEG should be done to identify nonconvulsive SE in those patients.
11. Nonepileptic spells simulating SE can be differentiated from SE by poorly coordinated thrashing, preserved consciousness or purposeful movements, head rolling, eyes held shut, back arching, and pelvic thrusting.

3.3.3 | Emergent treatment

The SC agreed to adopt the following recommendations:

1. Emergent prehospital treatment with benzodiazepines is needed; while assessing any abnormalities (hypoxia, hypoglycemia, or hypotension must be managed accordingly).
2. Suppose benzodiazepines are not administered to the patient before ED arrival, and the patient is still seizing. In

TABLE 2 Recommendations for choice of antiseizure medication (ASM) according to different epileptic syndromes in adolescents.

No.	Recommendations	Contraindicated medications
<i>Juvenile absence epilepsy (JAE)</i> ^{13,26,34}		
1.	First line: ethosuximide, lamotrigine, and sodium valproate ^a . Adjuvant therapy: a combination of any of the first lines. Other options: clobazam, clonazepam, levetiracetam, topiramate, and zonisamide.	Carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, tiagabine, and vigabatrin.
<i>Juvenile myoclonic epilepsy (JME)</i> ^{13,34}		
2.	First line: sodium valproate ^a and topiramate. Adjuvant therapy: acetazolamide, lamotrigine, levetiracetam, and topiramate. Other options: clobazam, clonazepam, and zonisamide.	Carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, tiagabine, and vigabatrin.
<i>Idiopathic generalized tonic-clonic seizures</i> ^{13,26,34,35}		
3.	First line: lamotrigine and sodium valproate ^a . Adjuvant therapy: consider carbamazepine and oxcarbazepine (exacerbating myoclonic and absence seizures). Other options: clobazam and topiramate.	-
<i>Other Idiopathic generalized seizures</i> ^{34–36}		
4.	First line; lamotrigine, sodium valproate ^a (especially if photosensitive), and topiramate. Adjuvant therapy: lamotrigine, levetiracetam, sodium valproate ^a , and topiramate. Other options: clobazam, clonazepam, and zonisamide.	Carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, tiagabine, and vigabatrin.

^aAvoid valproate in women of childbearing age.

that case, IV benzodiazepines should be included in the initial dosing when IV access is immediately available.

- When IV access is unavailable, IM, per rectum (PR), buccal, or intranasal benzodiazepines should be administered together with IV placement.
- Unless IV access is immediately available, initiate IM. In adults or children over 40 kg, IM midazolam is 10 mg. In children 13–40 kg, the IM midazolam dose is 5 mg; in children <13 kg, the IM midazolam dose is 0.2 mg/kg.
- When two IV accesses are immediately available, one for diazepam 0.15 mg/kg IV (up to 10 mg/dose) and the second line for one of the ASMs would be better to start simultaneously with the initial benzodiazepine rather than waiting for a response for benzodiazepine.
- Among second-line ASMs, the first-line choice will be either phenytoin or levetiracetam in their standard loading doses. Also, lacosamide can be used if the previous ASM failed or is unavailable.
- If seizures have stopped after the initial ASM and the patient has awakened, a maintenance dose of ASMs should be started if indicated. It can be given either orally or intravenously.

3.3.4 | Treatment of refractory status epilepticus

The SC agreed to adopt the following recommendations:

- SE is considered refractory when the seizures continue despite urgent and emergent treatment.
- Early (within 1 h) drug-induced coma with continuous IV infusion of an anesthetic drug is recommended for refractory status epilepticus (RSE) with EEG target of burst suppression.
- The recommended loading dose of IV midazolam^c infusion is 0.2 mg/kg at 2 mg/min. Then, repeated boluses every 5 min of 0.2–0.4 mg/kg should be administered until the seizures stop, up to a maximum loading dose of 2 mg/kg. Then, a continuous infusion should be started at 0.05–2 mg/kg/h.
- Propofol IV infusions are an alternative at 1–2 mg/kg IV over 3–5 min as a loading dose and then repeated

^cCheck midazolam availability before starting continuous infusion

TABLE 3 Recommendations for choice of antiseizure medication (ASM) according to seizure type.

No.	Recommendations	Avoid/do not offer
<i>Focal aware, focal with impaired awareness, and focal to bilateral tonic-clonic seizure</i> ^{14,34}		
1.	First-line monotherapy: lamotrigine or levetiracetam Second-line monotherapy: carbamazepine, oxcarbazepine, and zonisamide Third-line monotherapy: lacosamide First-line add-on/adjuvant; Carbamazepine, lacosamide, lamotrigine, levetiracetam, oxcarbazepine, topiramate, and zonisamide. Second-line add-on/adjuvant; brivaracetam, eslicarbazepine acetate, perampanel, pregabalin, and sodium valproate ^a . Third-line add-on; phenobarbital, phenytoin, tiagabine, and vigabatrin	-
<i>Generalized onset tonic-clonic seizure</i> ^{14,34}		
2.	First-line monotherapy; sodium valproate ^a lamotrigine and levetiracetam First-line add-on/adjuvant; lamotrigine, levetiracetam, sodium valproate ^a , topiramate, clobazam, and perampanel Second-line add-on; brivaracetam, lacosamide, phenobarbital, primidone, and zonisamide	-
<i>Generalized nonmotor (Absence) seizures</i> ^{14,34}		
3.	First-line monotherapy; ethosuximide Second-line monotherapy; sodium valproate ^a Third-line monotherapy; lamotrigine or levetiracetam If there is a high risk of other seizure types offer sodium valproate ^a or lamotrigine or levetiracetam as first line. Adjuvant/add-on; clonazepam, ethosuximide, lamotrigine, levetiracetam, sodium valproate ^a , topiramate, and zonisamide.	Carbamazepine, gabapentin, lacosamide, oxcarbazepine, phenytoin, pregabalin, and vigabatrin.
<i>Myoclonic seizures</i> ^{14,30,34}		
4.	First line: levetiracetam and sodium valproate ^a . Adjuvant/add-on: brivaracetam, clobazam, clonazepam, lamotrigine phenobarbital, piracetam, sodium valproate ^a , topiramate, and zonisamide.	Carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, and vigabatrin.
<i>Tonic and atonic seizures</i> ^{14,30}		
5.	First line: sodium valproate ^a and lamotrigine Adjuvant/add-on; lamotrigine, topiramate, clobazam, and rufinamide	Carbamazepine, gabapentin, oxcarbazepine, phenytoin, pregabalin, and vigabatrin.

^aAvoid valproate in women of childbearing age.

boluses every 3–5 min of the same amount until the seizures stop. The propofol infusion rate of 30–200 µg/kg/min should be maintained.

- Pentobarbital at 5 mg/kg as a loading dose, followed by a maintenance infusion of 1–3 mg/kg, may be used in children with refractory SE more frequently because of the adverse effects of propofol.
- Continuous EEG monitoring is mandatory during drug-induced coma and weaning after 24 h from starting the anesthetic drug. If not, we can perform EEG at least once every 24 h for 1 h while the patient is under the effect of anesthesia.
- At least a second ASM^d (phenytoin, sodium valproate, levetiracetam, lacosamide, and topiramate) must be added to the initial ASM in this stage before starting to wean the anesthetic drug.

^dChoices among different ASMs are determined according to comorbidities, availability, seizure types, and drug–drug interaction.

- Occurrence of seizures during weaning of anesthetic drug categorizes the patient as super-refractory SE (SRSE).

3.3.5 | Treatment of super-refractory status epilepticus

The SC agreed to adopt the following recommendations:

- Thiopental should be initiated in the recommended dose with progressive weaning of the previous anesthetic over the following 24 h.
- If a burst-suppression EEG pattern without epileptic activity has been achieved during 24 h under thiopental and SE recurs after thiopental weaning, adding phenobarbital should be considered in a therapeutic dose for at least 24 h, after which weaning should again be tried.
- If SRSE recurs after weaning of thiopental in association with phenobarbital or if it persists under thiopental

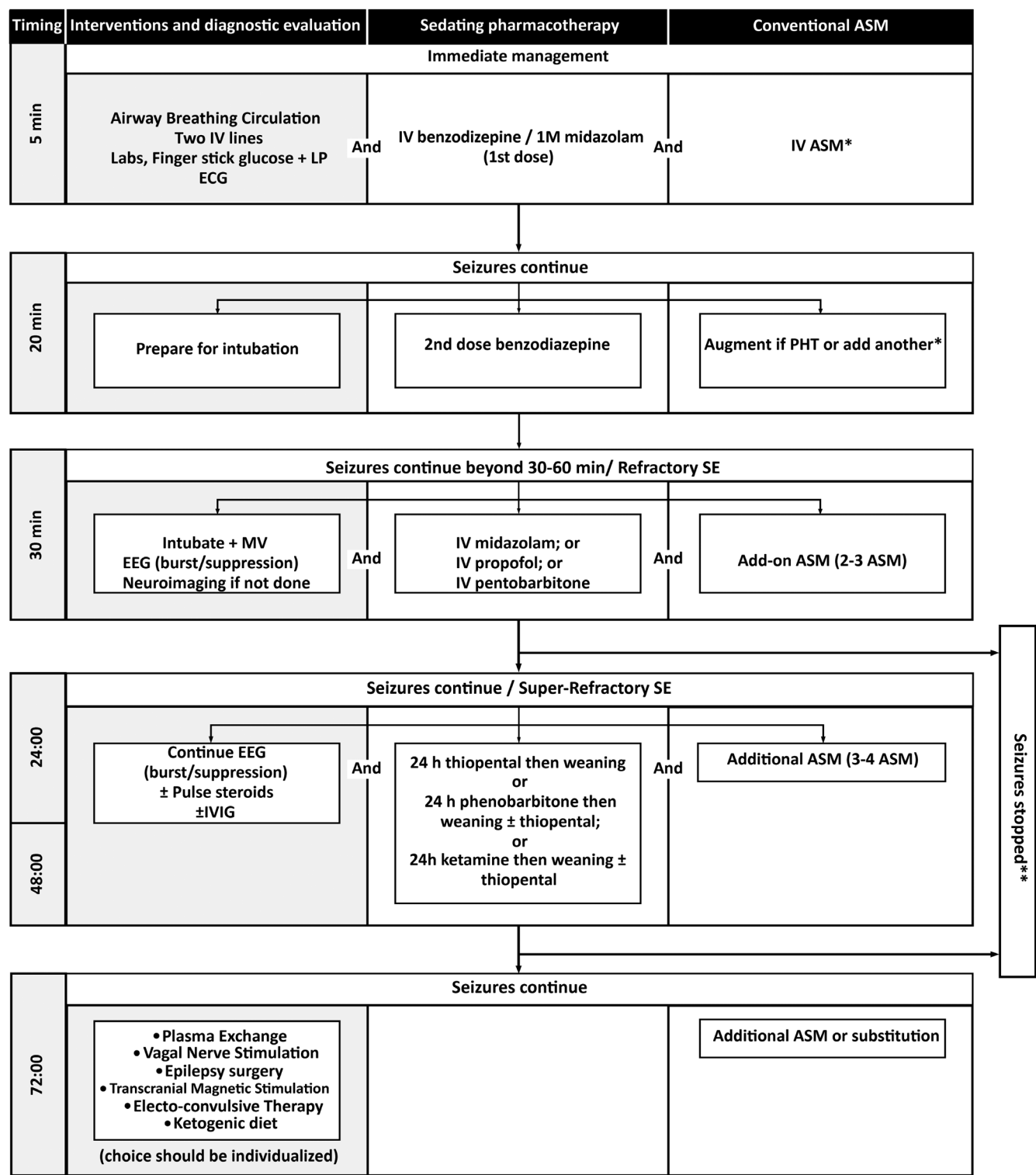


FIGURE 2 Status Epilepticus Management Algorithm. ASM, Antiseizure medication; ECG, electrocardiography; EEG, Electroencephalography; IM, Intramuscular; IV, Intravenous; IVIG, Intravenous immune globulin; LP, Lumbar puncture; MV, Mechanical ventilation; PHT, Phenytoin; SE, Status Epilepticus. *Phenytoin, levetiracetam, and lacosamide. **Gradual weaning of anesthetic and continue on ASM.

- anesthesia (epileptiform discharges), then ketamine in the recommended dose should be associated with thiopental.
4. If SRSE persists or recurs, progressively longer periods (>24 h) of sedation should be considered.
5. Each anesthetic should undergo gradual weaning—over 24 h—until complete suspension.
6. Simultaneous weaning of two anesthetics is not recommended.

7. In case of persistence or recurrence of SRSE despite the adoption of all previous recommendations, the patient should be maintained in burst suppression under combined anesthesia with thiopental and ketamine, and all of the following therapeutics should be considered (according to availability and applicability); magnesium sulfate—intravenous bolus (4g) followed by a continuous infusion (2–6 g/h) aiming for plasma levels of 3.5 mmol/L, pyridoxine 100–600 mg daily oral dose, immunotherapy, therapeutic hypothermia if available, and a ketogenic diet.
8. In case of persistence or recurrence of SRSE despite the adoption of all previous recommendations, the patient should be maintained in burst suppression under combined anesthesia with thiopental and ketamine, and all of the following therapeutics should be considered (according to availability and applicability); vagus nerve stimulation, epilepsy surgery, transcranial magnetic stimulation, trigeminal nerve stimulation, and electroconvulsive therapy

3.4 | Section IV: Drug-resistant epilepsy

More than 30% of patients with epilepsy still progress to DRE with a considerable increase in morbidity and mortality. The only possible way to cure patients with DRE may be surgical treatment. However, achieving a definitive effect with surgical treatment alone is always challenging. It needs to be combined with other treatment methods due to the complicated etiology and unclear pathogenesis of DRE.³⁴

3.4.1 | Definition

The SC agreed that the Egyptian epilepsy consensus adopts the ILAE definition of DRE as “failure of adequate trials of two or three tolerated, appropriately chosen and used antiseizure medication schedules (whether as monotherapies or in combination) to achieve sustained seizure freedom.”

3.4.2 | Who and when to refer for evaluation in a specialized/tertiary epilepsy care facility

The SC agreed to adopt the following recommendations:

1. All PWEs should access a tertiary service via their specialist whenever required.
2. When one or more of the following criteria are present, referral should be considered:

- (i) Uncontrolled epilepsy with medications within a maximum of 1 year.
- (ii) Management is unsuccessful after two adequate trials.
- (iii) Children <2 years of age (due to the profound psychological, developmental, and behavioral effects that might be correlated with continuing seizures).
- (iv) A PWE who is at risk of, or experiences, intolerable medication side effects.
- (v) Presence of a unilateral structural lesion in patients with DRE.
- (vi) Presence of a diagnostic doubt about the type of seizures and/or epilepsy syndrome.
- (vii) Children, young people, and adults with specific syndromes such as Sturge–Weber syndrome, hemispheric syndromes, Rasmussen's encephalitis, and hypothalamic hamartoma should be referred to a tertiary epilepsy service.
- (viii) Patients who require diet therapy for epilepsy.
- (ix) Psychiatric comorbidity and/or negative baseline investigations should not be a contraindication for referral to a tertiary service.

3.4.3 | Characteristics of a specialized/tertiary epilepsy facility

The median prevalence of epilepsy ranges between 4.9/1000 in developed countries and 5.9–12.7/1000 in developing countries.³⁵ In Arab countries, lifetime prevalence was reported to be 6.9/1000,³⁶ which is similar to an Egyptian study (6.98/1000) which also reported that 30.6% of these patients were uncontrolled.³⁷ In populous and growing countries such as Egypt (112464345), this indicates a parallel increase in epilepsy's burden that compels the establishment of specialized centers for adequate testing and appropriate treatments for PWE to improve quality of life. The SC agreed to adopt the following recommendations which specify the characteristics of epilepsy monitoring unit (EMU) as regards the healthcare team and the facilities required with the aim to encourage the founding of multiple EMUs that satisfy such imperative unmet needs

1. Multidisciplinary teams experienced in the assessment of patients with DRE should include psychology, psychiatry, neuroradiology, social worker, counseling, clinical nurse specialists, occupational therapy, neurology, neuroanesthesia, neurophysiology, and neurosurgery.
2. Investigations that must be available (i.e., as a minimal requirement) are as follows: video-EEG, MRI epilepsy

protocol, functional imaging, ASM drug level, and psychometric battery.

3. A specialized epilepsy facility should offer basic and advanced therapeutic intervention (s).

3.4.4 | Once referred to a specialized/tertiary epilepsy facility, the patient should undergo

The SC agreed to adopt the following recommendations:

1. History in detail (to exclude causes of pseudo-intractability): for example, PNES, cardiogenic (arrhythmias) and vasovagal events (syncope), parasomnias, movement disorders, seizure triggers, and drug-related problems: nonadherence, drug–drug interaction, or inappropriate choice or dose.
2. Review and order investigations, including routine labs such as CBC, liver and kidney function tests, electrolytes (calcium, sodium, and magnesium), blood sugar, screening metabolic studies, and genetic epilepsy panel.
3. Video-EEG to characterize the epilepsy types and rule out PNES.
4. Imaging (MRI-HARNESS protocol) studies must be scrutinized once epilepsy diagnosis is confirmed to ascertain the presence of an epileptogenic focus and facilitate a possible surgical work-up.
5. In the presence of high suspicion of focal seizures, if no epileptogenic lesion is found on MRI, other ancillary tests, including ictal single-photon emission computed tomography (SPECT) and positron emission tomography (PET), should be performed. In addition, eloquent regions of the brain can be studied using functional studies such as functional MRI.
6. Invasive testing can be considered in case of unclear localization or if a more precise definition of the relationship of the eloquent cortex to the epileptic cortex is needed.
7. Psychiatric and psychometric testing are usually warranted in the patients' presurgical evaluation.

3.4.5 | Epilepsy surgery referrals in Egypt

The SC agreed to adopt the following recommendations (Figure 3): Epilepsy Surgery Algorithm.

2. Sustained Seizure Freedom: Seizure-free duration at least three times the longest interseizure interval prior to starting a new intervention (determined from seizures occurring within the past 12 months) or 12 months, whichever is longer.
3. Treatment Failure: Recurrent seizure(s) after the intervention has been adequately applied.
4. Undetermined Seizure-Free Outcome: If a patient has been seizure-free three times the preintervention interseizure interval but less than 12 months.
5. Patients who fulfill the criteria (regardless of age) for DRE should be referred to a Specialized Epilepsy Center (SEC) for reassessment to nonpharmacological management.
6. Specialized Epilepsy Center consists of an epilepsy monitoring unit (EMU), run by an epileptologist/neurophysiologist and functional neurosurgeon with infrastructure and a multidisciplinary team of health-care professionals (psychologist, EEG technician, nurses, medical secretary, and medical registry).
7. The Egyptian Epilepsy consensus recommends using the modified CASES Expert Panelists online tool³⁸ to identify patients who may benefit from an epilepsy surgery evaluation.
8. The Egyptian epilepsy consensus recommends using the criteria for referral and evaluation of children for epilepsy surgery proposed by the Sub-Commission for Pediatric Epilepsy Surgery of the ILAE.
9. Children with uncontrolled (i.e., failure of two or three appropriate drugs) and disabling epilepsy are also possible surgical candidates.
10. Children with lateralized seizures or other evidence of focality that cannot be attributed to idiopathic focal epilepsies or in whom the MRI reveals a lesion amenable to surgical removal should be referred to a pediatric SEC for evaluation.
11. In the pediatric surgical population, there are currently no preoperative clinical variables to predict seizure outcome; therefore, the presence of developmental delay, physical, and/or psychiatric comorbidities should not be a contraindication for pediatric epilepsy surgery and/or neuromodulation (e.g., vagal nerve stimulation “VNS”).
12. The aim of testing for epilepsy surgery is to determine if a person can benefit from surgery to treat their seizures.
13. Presurgical evaluation tests are done in two phases. Phase 1 can be done in the outpatient setting. While in phase 2 tests, a hospital stay will be required.
14. The number of presurgical evaluation tests depends on the person's type of seizures and the information needed to make the best decisions about moving forward with the presurgical evaluation and surgery.

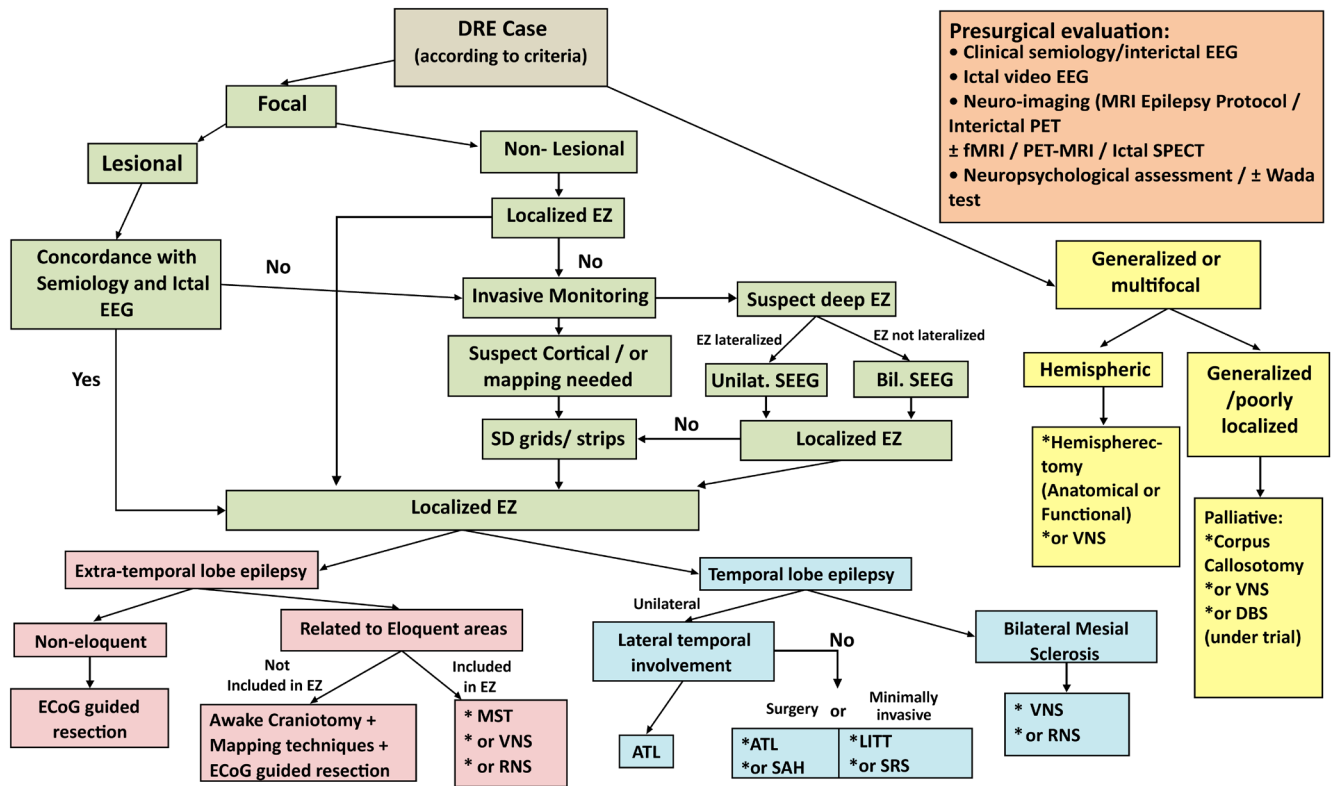


FIGURE 3 Epilepsy Surgery Algorithm. ATL, anterior temporal lobectomy; DBS, deep brain stimulation; DRE, drug-resistant epilepsy; ECoG, electrocorticography; EZ, epileptogenic zone; LITT, laser interstitial thermal therapy; MST, multiple subpial transections; RNS, responsive neurostimulation; SAH, selective amygdalohippocampectomy; SD grids, subdural grids; SEEG, stereotactic electroencephalography; SRS, stereotactic radiosurgery; VNS, vagal nerve stimulation.

15. Phase 1 (noninvasive) tests should include any of the following as required:
 - a. EEG
 - b. Continuous video-EEG monitoring
 - c. High-density EEG and source localization—OPTIONAL
 - d. Special electrodes—OPTIONAL
 - e. CT scan without/with IV contrast
 - f. MRI without/with IV contrast
 - g. PET—OPTIONAL
 - h. MEG (magnetoencephalography)—OPTIONAL
 - i. SPECT—OPTIONAL
 - j. SISCOM (subtraction ictal SPECT co-registered with MRI)—OPTIONAL
 - k. MRS (magnetic resonance spectroscopy)—OPTIONAL
 - l. Neuropsychology testing—REQUIRED
 - m. Wada test^{*39}
16. When imaging is used as part of the presurgical evaluation, strict standards for technique and interpretation must be used.
17. Imaging results must be placed in the context of other clinical and laboratory data, including clinical history, neurologic examination, video recordings of typical seizures, ictal and interictal EEG, and neuropsychometric evaluation.
18. Images should be reviewed by physicians experienced in evaluating patients with epilepsy and neuroradiologists.
19. Phase 2 tests include:
 - a. Intracranial Depth electrodes (also called stereo-EEG or SEEG)
 - b. Subdural strip and subdural grid electrodes
 - c. ECoG (electrocorticography) or iEEG (intracranial electroencephalography)
20. Surgical decision making is a multidisciplinary decision of the entire epilepsy team.
21. Presurgical evaluation should include:
 - a. Clinical semiology/interictal EEG
 - b. Ictal video-EEG
 - c. Neuroimaging: MRI epilepsy protocol includes 3D FLAIR and volumetric studies
 - d. Neuroimaging: Interictal PET: for identifying the hypometabolic epileptogenic zone.
 - e. Neuropsychometric assessment

*For evaluating the role of WT as a presurgical test, please refer to systematic review by Andreu Massot-Tarrús and Seyed M. Mirsattar

22. Focal epilepsy:

- a. Focal epilepsy should be classified into lesional or nonlesional subclasses based on the results of the MRI epilepsy protocol.
- b. Invasive monitoring should be applied for lesional focal epilepsy, not concordant with semiology and EEG unless the lesion requires surgical resection for another reason such as low-grade glioma.
- c. Consider invasive monitoring should be applied for nonlesional focal epilepsy with a nonlocalized epileptogenic zone.*
- d. If a cortical epileptogenic zone is suspected/or mapping is needed, intracranial EEG monitoring should be applied.
- e. If a deep epileptogenic zone is suspected, alternative and more invasive monitoring approach could be applied according to institutional preferences and experience such as sEEG, strips, and grids.
- f. In the case of the localized epileptogenic zone, proceeding to direct surgical approach was defined in view of previous recommendations and in consideration with the skip criteria that allow for direct surgical resection in focal epilepsy such as absence of bilateral temporal lobe epilepsy, extratemporal lobe epilepsy with eloquent areas included in the epileptogenic zone, or dominant temporal lobe epilepsy, in which VNS, responsive neurostimulation (RNS), and multiple subpial transections (MST) could be an alternative. Agreement for surgery in lesional focal epilepsy in concordance with semiology and EEG, or nonlesional focal epilepsy with the localized epileptogenic zone, was outlined as follows:
 - (i) For noneloquent extra-temporal lobe epilepsy, ECoG-guided resection can be considered
 - (ii) Intraoperative or perioperative mapping of functional localization and or seizure onset zone can be used for extra-temporal lobe epilepsy related to eloquent areas but not included in the epileptogenic zone
 - (iii) For extra-temporal lobe epilepsy with eloquent areas included in the epileptogenic zone, implementation of procedures such as MST, VNS, or RNS should be applied.
 - (iv) For dominant temporal lobe epilepsy, implementation of procedures such as MST, VNS, or RNS could be applied
 - (v) For unilateral temporal lobe epilepsy with lateral temporal involvement, anterior temporal lobectomy (ATL) should be applied.

- (vi) For unilateral temporal lobe epilepsy with no lateral temporal involvement, either surgical procedures (ATL or selective amygdalohippocampectomy “SAH”) or minimally invasive procedures (laser interstitial thermal therapy “LITT”) could be implemented. Stereotactic radiosurgery “SRS” and radiofrequency ablation are alternatives of doubtful outcomes and should not be applied unless other effective surgical/LITT is not indicated, failed, or unavailable. This also applies to DBS and RNS, which can be an option in patients with unilateral temporal lobe epilepsy who are at cognitive risk or wish to not pursue any cognitive risk.
- (vii) For bilateral temporal lobe epilepsy, VNS, RNS, or DBS could be applied. It is worth mentioning that RNS is not available outside the USA, and it is an option only when it becomes available and approved by the country's health authorities.

23. Generalized or multifocal epilepsy:

- a. In the case of hemispheric generalized or multifocal epilepsy, hemispherectomy (anatomical or functional), VNS, or deep brain stimulation (DBS) could be applied.
- b. In the case of generalized, poorly localized epilepsy with mostly atonic attacks, palliative management with corpus callosotomy could be applied.
- c. In the case of generalized poorly localized epilepsy with minor atonic attacks, palliative management with VNS or DBS of the anterior nucleus of the thalamus should be applied.

4 | DISCUSSION

Epilepsy is one of the most common chronic diseases in the neurological field, estimated to affect approximately 68 million people worldwide. The epidemiologic reports describing the burden of epilepsy in the last years are problematic because epilepsy is still scarce in many parts of the world, with differences in sampling, definitions, measurements, screening, diagnostic accuracy, and different management approaches.⁴⁰ Proper and adequate management strategies of DRE have critical reflection and favorable impact on the clinical practice through proper guidelines/consensus as an identified key value driver. However, despite the availability of general epilepsy guidelines, there were no specific guidelines for DRE in Egypt, with an urgent need for a practical pathway to focal DRE and epilepsy surgery, especially the urgent need for the availability of a comprehensive specialized epilepsy center for proper epilepsy management.

*Some patients could be candidates for surgery without PET if the epileptogenic zone is concordantly localized by semiology, ictal scalp EEG, and MRI

The goal of the management strategy in patients with epileptic seizures is to achieve a seizure-free outcome with maximum safety. This goal is accomplished in two-thirds of patients requiring treatment with ASMs. However, many patients experience adverse effects from ASMs, and some patients have seizures that are pharmacoresistant and may require further surgical approach strategy.⁴¹

The prevalence is reported to vary between developed and limited-resource countries: estimated as 4–7 per 1000 persons in developed countries and 5–74 per 1000 persons in resource-limited ones. The wider variations in prevalence complicate the use of these data in estimating the number of epileptic patients who may benefit from treatment and in informing health authority policy.⁴⁰

The previously reported and impressive disparities in resources for epilepsy care in developed and resource-limited countries have been clearly demonstrated in two WHO projects. In many resource-limited countries, epilepsy care is relegated to the mental health sections, which are the most underfunded health system.⁴²

To improve epilepsy care in any limited-resource region, the goal should not be to replicate guidelines, systems, and processes of epilepsy care from developed countries in limited-resource countries. Care must be cost-effective, and appropriately provided culturally and socially, reflecting the society's limited resources and higher priorities. Systems and epilepsy care processes must be individualized.⁴³

The epilepsy gap could be higher in a limited-resource country such as Egypt because of a lack of multidisciplinary experience in the public health sector. The epilepsy treatment gap is defined by the International League Against Epilepsy as follows: “The difference between the number of people with active epilepsy and the number whose seizures are being appropriately treated in a given population at a given time expressed as a percentage. This definition includes diagnostic and therapeutic deficits.”⁴⁴

Our aim was to minimize such a huge epilepsy gap in Egypt by working to adapt, uniform, and guide the epilepsy care process in Egypt by creating and propagating such consensus through healthcare providers and coordinating the work between healthcare providers and the health authority in the country. We designed a competency-based approach through several meetings with epilepsy care experts from different universities and health institutes covering broad geographical areas in Egypt. Several rounds were well organized and controlled to have the feedback with efforts to minimize all sources of bias, such as preventing group interaction and anonymous process. Finally, we created our adapted and well-constructed consensus to be propagated through all health care and epilepsy care providers under coordination between all health authorities and insurance systems.

Through the study, we identified several needs, including the insufficient number of referrals to specialized epilepsy units that lead to misdiagnosis, late detection, and delayed effective therapies. Another key issue referred to the lack of effective pharmacological management and suboptimal referrals to specialized epilepsy surgery centers. All participants agreed that DRE is a critical condition concerning the need for more effective ASMs with the potential to change the natural history of DRE and the need for an earlier, effective, adequate, and proper diagnosis to improve epilepsy management. It was figured out that, in a clinical scenario, there is inequity in the availability of resources to treat DRE across Egyptian centers (e.g., video-EEG evaluation, frequency of medical visits, and expertise of healthcare professionals). We also noted that the use of medications is suboptimal, providing only symptomatic relief and not impacting the course of the disease since they are limited to reducing the number of seizures, and seizure freedom is not achieved in a significant proportion of patients. Similar gap is noted regarding referrals to specialized epilepsy surgery centers despite its availability. This may be due to the current lack of national guidelines or unclear referral conditions. By the release of this consensus, such gaps will be minimized.

Our consensus perspective also aims to minimize the burden on the health authority and community by reducing the malpractice in epilepsy care. Several unfavorable factors could be minimized through propagating our consensus, such as minimizing unnecessary hospital stays, visits to emergency units, outpatient procedures, and implementation of epilepsy surgery when indicated. Established epilepsy protocols including implementation of epilepsy surgery and well-oriented healthcare providers, especially in a limited-resource country, will dramatically reduce the significant financial burden on patients, their families, and caregivers driven by associated lack of productivity, cognitive dysfunction, unemployment, and early retirement.

Of course, these proposed recommendations have certain limitations that should be clarified. First, we used the Delphi process for expert consensus generation, rather than evidence-based guidelines, which require a high grade of supporting evidence that is not available in resource-limited countries; however, thorough review of previous guidelines was done to strengthen the credibility of our recommendations.

Second, although not all relevant stakeholders were included as paramedical staff/nurses and patients, the consensus party was formed of all professional subgroups: neurologists, epileptologists, neurophysiologists, functional neurosurgeons, and neurologists with additional specialty in neuropsychology and neuropsychiatric. It is intended to add any relevant groups as paramedical staff, radiologists, and patient perspectives, especially after

applying our new guidelines to upcoming updates after applying our new guidelines.

Lastly, not all issues regarding epilepsy management are included, consensus regarding management of special populations as women with epilepsy (WWE) is intended to be considered separately. The consensus mentioned specified issues related to women when indicated; however, details of management regarding sexual dysfunction, fertility, contraception, catamenial seizures, pregnancy, teratogenicity of ASMs, safety of ASMs during lactation, and menopause will be addressed in an upcoming publication.

AUTHOR CONTRIBUTIONS

All authors listed have made a substantial, direct, and intellectual contribution to the work. The final manuscript draft was reviewed and approved by all authors before publication.

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All authors listed have no conflict of interest, financial, or otherwise.

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The datasets used and/or analyzed during the current investigation are available upon reasonable request from the corresponding author.

ETHICS STATEMENT

Not applicable.

CONSENT FOR PUBLICATION

All authors have approved this paper's contents. "We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines."

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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